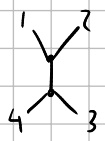
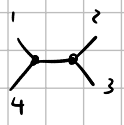
$A_4 =$ $a_1$  $+$ $a_2$ 

$s_{12}$ $s_{13}$ $s_{14}$ $s_{23}$ $s_{24}$ $s_{34}$

impose the hidden zero condition $\Rightarrow$ $a_1 = a_2$

$$\frac{1}{x}(\cdots) + \frac{1}{y}(\cdots)$$

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<u>goal</u>: show that some of the $X_{ij}$ variables satisfy the same relations if we set a 1-zero <u>or</u> perform a BCFW shift (up to order $z$)

<u>1-zero</u>: $c_{13} = c_{14} = \cdots = c_{1n-1} = 0$ $\rightarrow$ $n-3$ constraints

started with $\frac{n(n-3)}{2}$ variables, now we have $n-3$ less of them

we know that $c_{ij} = X_{ij} + X_{i+1\,j+1} - X_{i+1\,j} - X_{i\,j+1}$

$$0 = X_{13} + X_{24} - \underbrace{X_{23}}_{=0} - X_{14} \qquad \text{since } c_{13} = 0$$

$$0 = X_{14} + X_{25} - X_{24} - X_{15}$$

$$\vdots$$

$$0 = X_{1n-1} + X_{2n} - X_{2n-1} - \underbrace{X_{1n}}_{=0}$$

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = X_{24} - X_{14} + X_{15} = X_{15} - X_{13}$$

$$X_{26} = X_{16} - X_{13}$$

$$\vdots$$

$$X_{2n} = X_{1n} - X_{13} = -X_{13}$$

$X_{21} = X_{23} = 0$

let's do shift instead $\quad P_2 \to P_2 + q z$

$\qquad\qquad\qquad\qquad\qquad\qquad P_n \to P_n - q z$

which $X_{ij}$ are effected by the shift? (assume $i<j$)

$$X_{ij} = (P_i + P_{i+1} + \cdots + P_{j-1})^2$$

$i=1$ or $i=2$ $\qquad X_{1j} \quad j = 3, \ldots, n-1$ $\qquad$ are the only effected ones

$\qquad\qquad\qquad\qquad X_{2j} \quad j = 4, \ldots, n$

expand $X_{1j}(z)$ to linear order

$\qquad X_{2j}(z)$ to linear order

$$X_{1j}(z) = (P_1 + P_2 + qz + \cdots + P_{j-1})^2$$

$$X_{2j}(z) = (P_2 + qz + \cdots + P_{j-1})^2$$

$$X_{1j}(z) = (P_1 + P_2 + qz + \cdots + P_{j-1})^2$$

$$= (\underbrace{qz} + \underbrace{P_1 + P_2 + \cdots P_{j-1}})^2$$

$$= (P_1 + P_2 \cdots + P_{j-1})^2 + z^2 \cancel{q^2}^{=0} + 2zq \cdot (P_1 + P_2 + \cdots + P_{j-1})$$

$$= X_{1j}(0) + 2zq \cdot (P_1 + P_3 + \cdots P_{j-1}) \quad \to \text{call it } X_{1j}^{(\infty)}$$

$$= X_{1j}(0) + X_{1j}^{(\infty)} z$$

$z \sim 1 + z$

$$X_{2j}(z) = (P_2 + qz + \cdots + P_{j-1})^2$$

$$= X_{2j}(0) + 2qz \cdot (P_3 + \cdots + P_{j-1}) \quad \to X_{2j}^{(\infty)}$$

$$= X_{2j}(0) + X_{2j}^{(\infty)} z$$

$$X_{2j}^{(\infty)} = 2q \cdot (P_3 + \dots + P_{j-1})$$

$$X_{1j}^{(\infty)} = 2q \cdot (P_1 + P_3 + \dots + P_{j-1})$$

$$= \underbrace{2q \cdot P_1}_{= X_{13}^{(\infty)}} + X_{2j}^{(\infty)}$$

$$X_{2j}^{(\infty)} = X_{1j}^{(\infty)} - X_{13}^{(\infty)}$$

$$X_{2j}^{(\infty)} = X_{1j}^{(\infty)} - X_{13}^{(\infty)}$$

$$X_{24}^{(0)} = X_{14}^{(0)} - X_{13}^{(0)}$$
$$X_{25}^{(0)} = X_{15}^{(0)} - X_{13}^{(0)}$$
$$X_{26}^{(0)} = X_{16}^{(0)} - X_{13}^{(0)}$$
$$\vdots$$
$$X_{2n}^{(0)} = X_{1n}^{(0)} - X_{13}^{(0)} = - X_{13}^{(0)}$$

---

let $B$ be an homogeneous rational function of the $X_{ij}$

e function $f(\lambda x_1, \lambda x_2, \dots, \lambda x_n) = \lambda^n f(x_1, \dots, x_n)$

let $B_i$ be the subset of all terms in $B$ that under the shift $P_2 \to P_2 + 2q$ $\quad P_n \to P_n - 2q$ scales as $z^i$ as $z \to \infty$

$i \in \mathbb{Z}$

<u>ex:</u>

$$B(z) = \frac{1}{X_{24}(z)} = \frac{1}{X_{2j}(0) + X_{2j}^{(\infty)} z}$$ scales like $\frac{1}{z}$

$\to B_{-1}$

$$B(z) = \underbrace{X_{25}(z) + X_{26}(z)}_{B_1(z)} + \underbrace{\frac{X_{35}^2}{X_{24}(z)}}_{B_{-1}(z)} + \underbrace{X_{35}}_{B_0(z)}$$

in general: $B = B_m + B_{m-1} + B_{m-2}$ where $z^m$ is the largest scaling

**Prop:** these are equivalent:

① • $B$ satisfies the 1-zero $\overset{\text{def}}{\Longleftrightarrow}$ ( when $X_{2j} = X_{1j} - X_{13}$ holds, $B = 0$ )

② • each $B_i$ satisfies the 1-zero

③ • each $B_i$ has enhanced UV scaling $z^{i-1}$

$z - z$ has UV scaling $z^0$

**proof** suppose ① holds

the only way this can happen

→ must be $B = \sum_{c_{ij} \in \text{zero}} c_{ij} A_{ij}$ some function of $X_{ij}$

$(c_{13}, c_{14}, \dots c_{1n-1})$

ex: $B = X_{24} - X_{13} + X_{14}$

now let's look at scaling

as $z \to \infty$ $\quad B = z^m B_m(X_{ij}^{(\infty)}) + O(z^{m-1})$

ex: $B(z) = \dfrac{1}{X_{24}(z)} = \dfrac{1}{X_{2j}(0) + X_{2j}^{(\infty)} z}$

$B(z) \to \dfrac{1}{X_{2j}^{(\infty)}} \dfrac{1}{z} = z^{-1} B_{-1}(X_{2j}^{\infty})$

if $B$ has enhanced scaling then

$B_m(X_{ij}^{\infty}) = 0$

↳ the only solutions they satisfy are $X_{2j}^{\infty} = X_{1j}^{\infty} - X_{13}^{\infty}$

$B_m$ satisfies the k-zero

so $B_m = \sum_{c_{ij} \in \text{zero}} c_{ij} \{\dots\}_{ij}$

$$X_{2j}(0) = X_{1j}(0) - X_{13}(0) \qquad\qquad X_{2j} = X_{2j}(0) + X_{2j}^{\infty} z$$

$$\Updownarrow$$

$$X_{2j}(z) = X_{1j}(z) - X_{13}(z)$$

$B$ satisfies 1-zero then

satisfy

as $z \to \infty$ $\quad B(X_{ij}(z)) = 0$

$$\| \quad \underbrace{B_m(X_{ij}^{\infty})}_{\to 0} z^m + O(z^{m-1}) \qquad B_m(X_{ij}^{\infty}) = 0$$

$B$ satisfies 1-zero $\Rightarrow$ enhanced scaling of $B_m$ $\Rightarrow$ $B_m$ satisfies 1-zero

$n = 4$ $\qquad B = \dfrac{1}{X_{13}} + \dfrac{1}{X_{24}}$ $\qquad$ set $\quad c_{13} = 0$

$$\Updownarrow$$

$$X_{24} = -X_{13}$$

$$B = \frac{1}{X_{13}}$$

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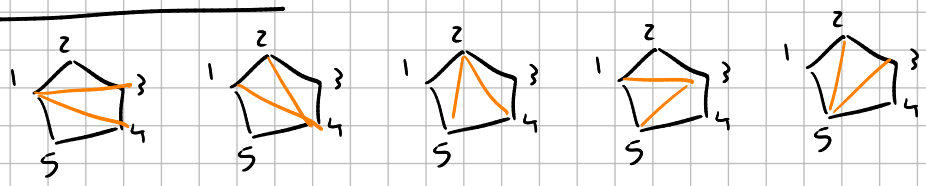
$n = 5$




Suppose

$$B = \underbrace{\frac{a_1}{X_{13}X_{14}} + \frac{a_2}{X_{24}X_{14}} + \frac{a_3}{X_{24}X_{25}}}_{B_{-2}} + \underbrace{\frac{b_1}{X_{13}X_{35}} + \frac{b_2}{X_{25}X_{35}}}_{B_{-1}}$$

$z^{-2}$ $\qquad z^{-1}$

require that $B$ satisfies the 1-zero

---

the only term that has $\dfrac{1}{X_{35}}$ is $\dfrac{b_1}{X_{13}X_{35}} + \dfrac{b_2}{X_{25}X_{35}}$

$$0 = B(X^{(0)} \text{ and } X_{35} = 1) = B_{-2} + \frac{b_1}{X_{13}} - \frac{b_2}{X_{13}}$$

$$0 = B\left(X^{(n)} \text{ and } X_{35} = \tfrac{1}{2}\right) = B_{-2} + 2\frac{b_1}{x_{13}} - 2\frac{b_2}{x_{13}} \quad \forall \; x_{ij}$$

$$\Rightarrow B_{-2}(X^{(n)}) = 0 \quad \text{and} \quad \frac{b_1}{x_{13}} - \frac{b_2}{x_{13}} = 0 \quad \rightarrow b_2 = b_1$$

$$\Downarrow$$